# AI-Powered Multilingual Reports with Azure AI and Semantic Link Labs

Samantha Cruz

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# Agenda Slide

1. Introduction
2. When do we need multilingual reports?
3. Components of a Report
4. Translation Strategy
5. Tools
6. Best Practices



SPEAKER

# Samantha Cruz

Senior Data Engineer at isolutions
Technical Instructor at Databricks
Certified: 4x Databricks 3x Microsoft
Also worked at Glovo, AB-Inbev, APCOA Parking

LinkedIn

**“68% of business leaders say miscommunication between languages affects productivity and decision-making”**

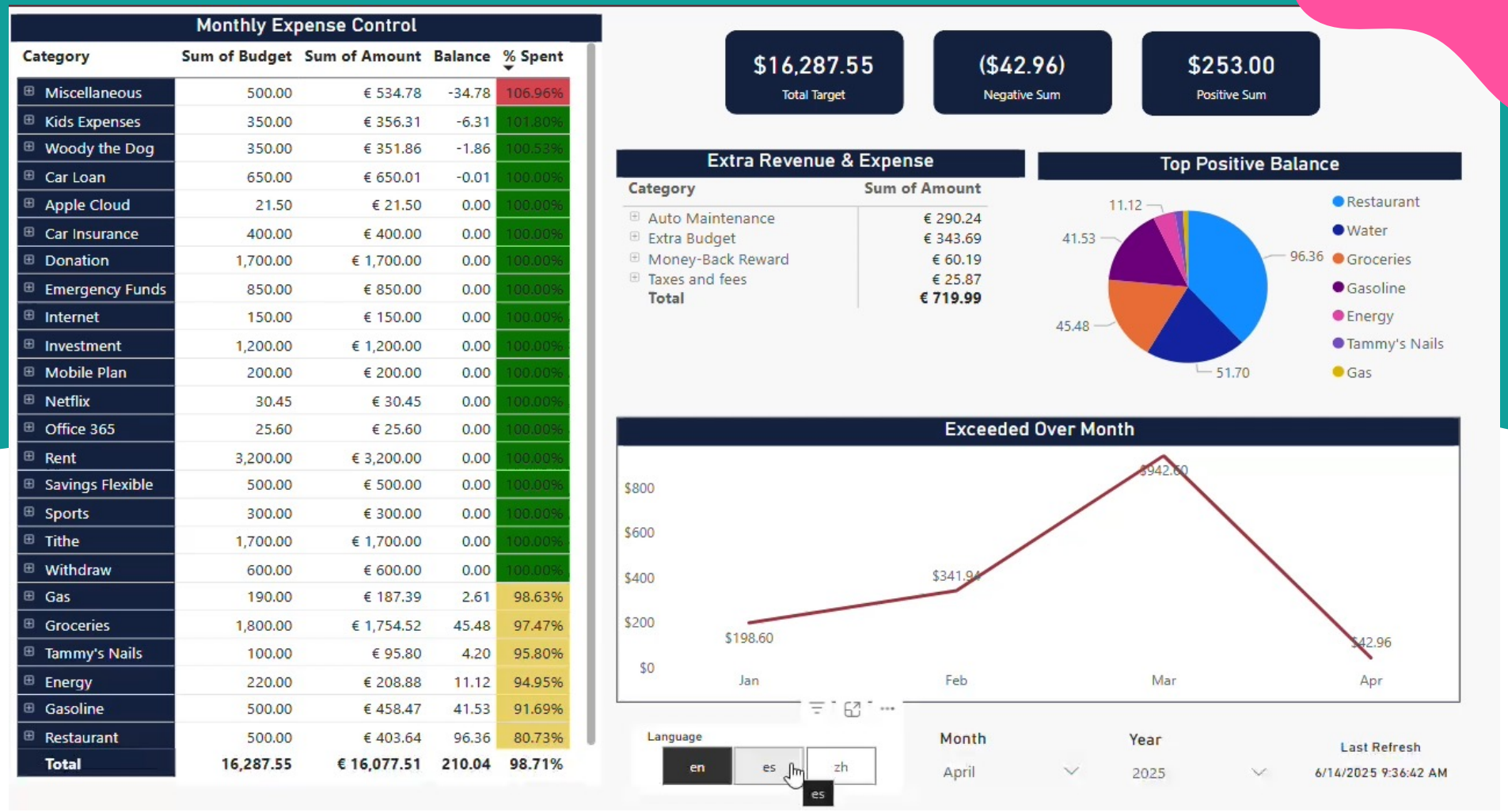
***Deepl Research***

**“90% of modern workforces struggle with language barriers; 67% say miscommunications lead to inefficiency”**

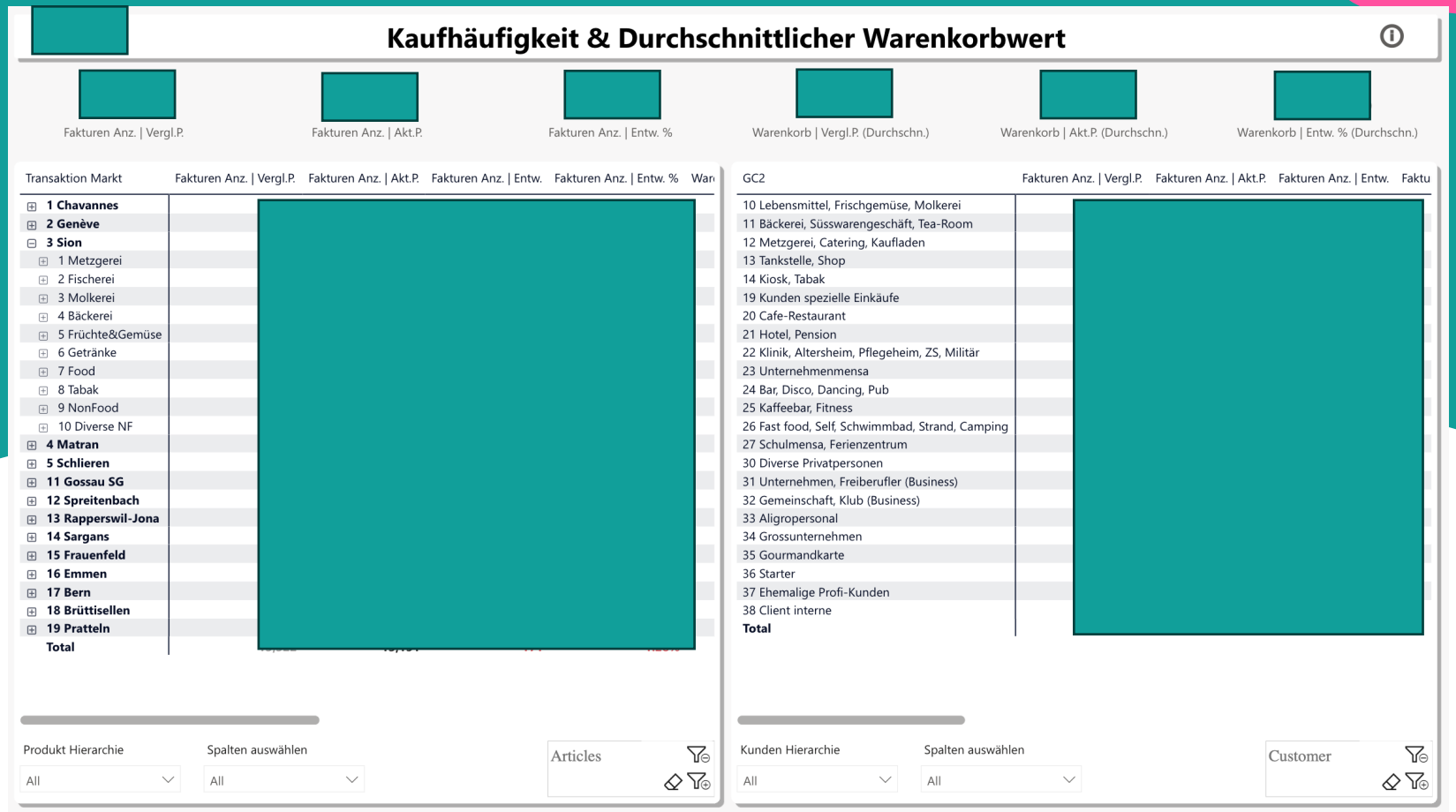
***Forbes Insights***

Sources: https://www.deepl.com/en/blog/us-executive-deepl-research
https://images.forbes.com/forbesinsights/StudyPDFs/Rosetta_Stone_Report.pdf

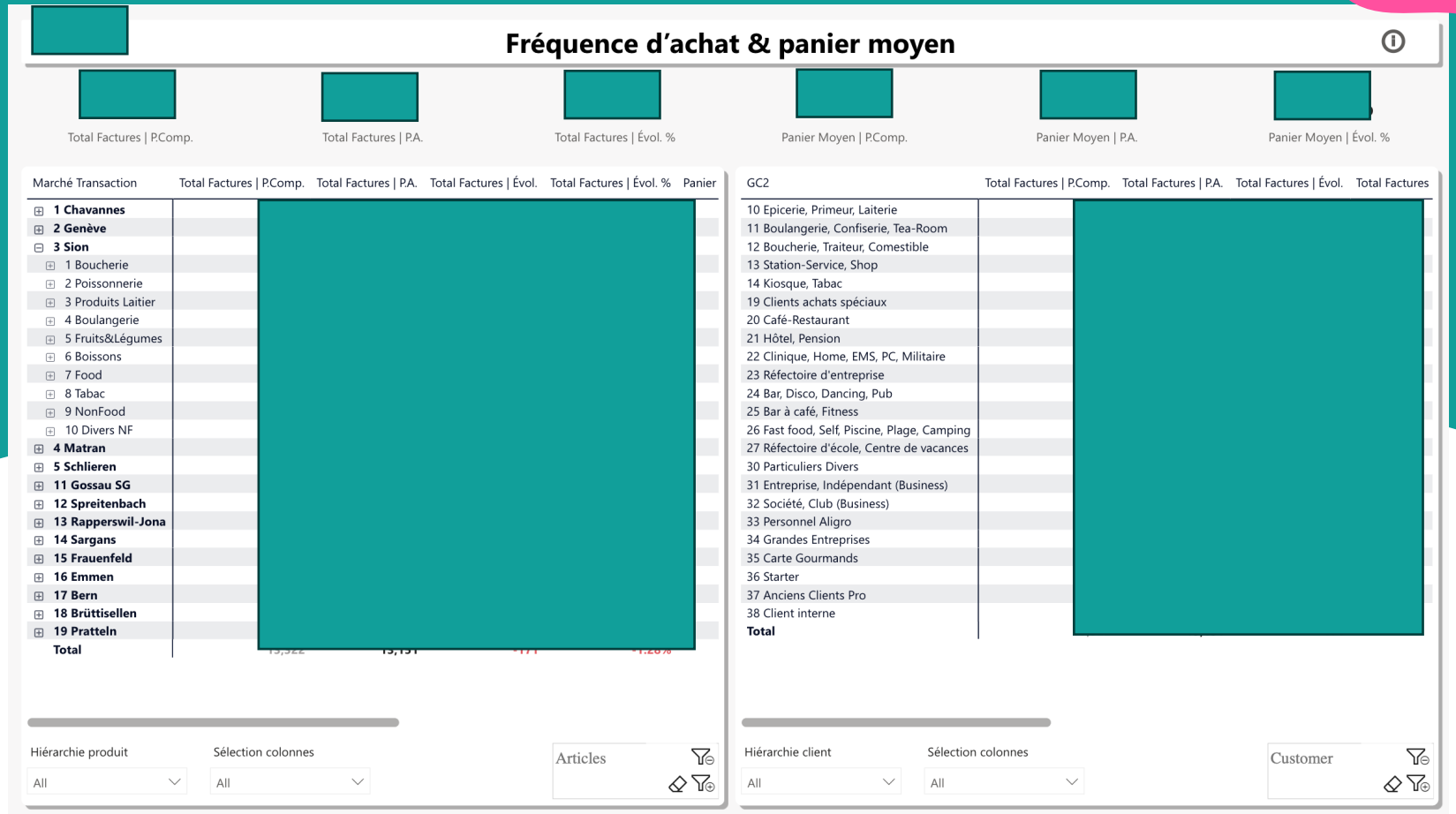
# Can we make a report fully multilingual?



# Can we make a report fully multilingual?



# Can we make a report fully multilingual?



# When do we need multilingual reports?

- International companies with global teams
- Regional managers in different countries
- Government and public sector reports
- Customer-facing dashboards
- Legal or compliance reporting
- Multi-brand/multi-country platforms
- Training and onboarding materials
- Accessibility and inclusion initiatives

# Understanding the Components of a Report

1. **Data Layer (Content stored in the model)**
   - Table names
   - Column names
   - Measures
   - Data values (text fields)
2. **Metadata Layer (How things are labeled or structured)**
   - Renamed fields
   - Tooltips
   - Field display names
   - Titles of visuals
3. **Report Layer (UI & user-facing content)**
   - Buttons
   - Text boxes
   - Page descriptions
   - Static labels and annotations

# Understanding the Components of a Report

| Area | End users | Developers / Self-Service | How (very brief) |
|---|---|---|---|
| **Metadata translation** (table/column/measure captions, descriptions, folders, hierarchies, perspectives, calc group item names, Q&A synonyms) | **Optional** — only if users can *Personalize visuals*, use *Analyze in Excel*, or browse the *Field list*. | **Must** — ensures localized field list, tooltips, folders, and consistency across self-service and reused datasets. | Tabular Editor → **Manage Translations** (add e.g., de-DE, en-US) → fill captions/descriptions/folders/hierarchies. |
| **Report label translation** (visual titles, subtitles, axis titles, legends, button labels, card/KPI labels, tooltip text, page titles/nav, disclaimers, alt text) | **Must** — that's what end users read inside the report. | **Must** (for any shipped report) — even devs need these wired to test UX. | Translation table + **DAX measures** (CALCULATE/MAX by Key & Language). Page titles via **textbox + measure**. Optional **dynamic format strings** for currency/number/date. |
| **Data translation** (dimension members shown to users: categories, statuses, month/week names, Yes/No) | **Must if displayed** — otherwise users see the wrong language; ensure correct **Sort by** (e.g., MonthNum). | **Should** — if the dataset/report is used in multiple regions or reused; **Not needed** if values are codes or one language only. | Either **dual columns + Field Parameter** (e.g., Month_DE/Month_EN) or a **normalized translation table** joined on keys. Keep a numeric **Sort by** column. |

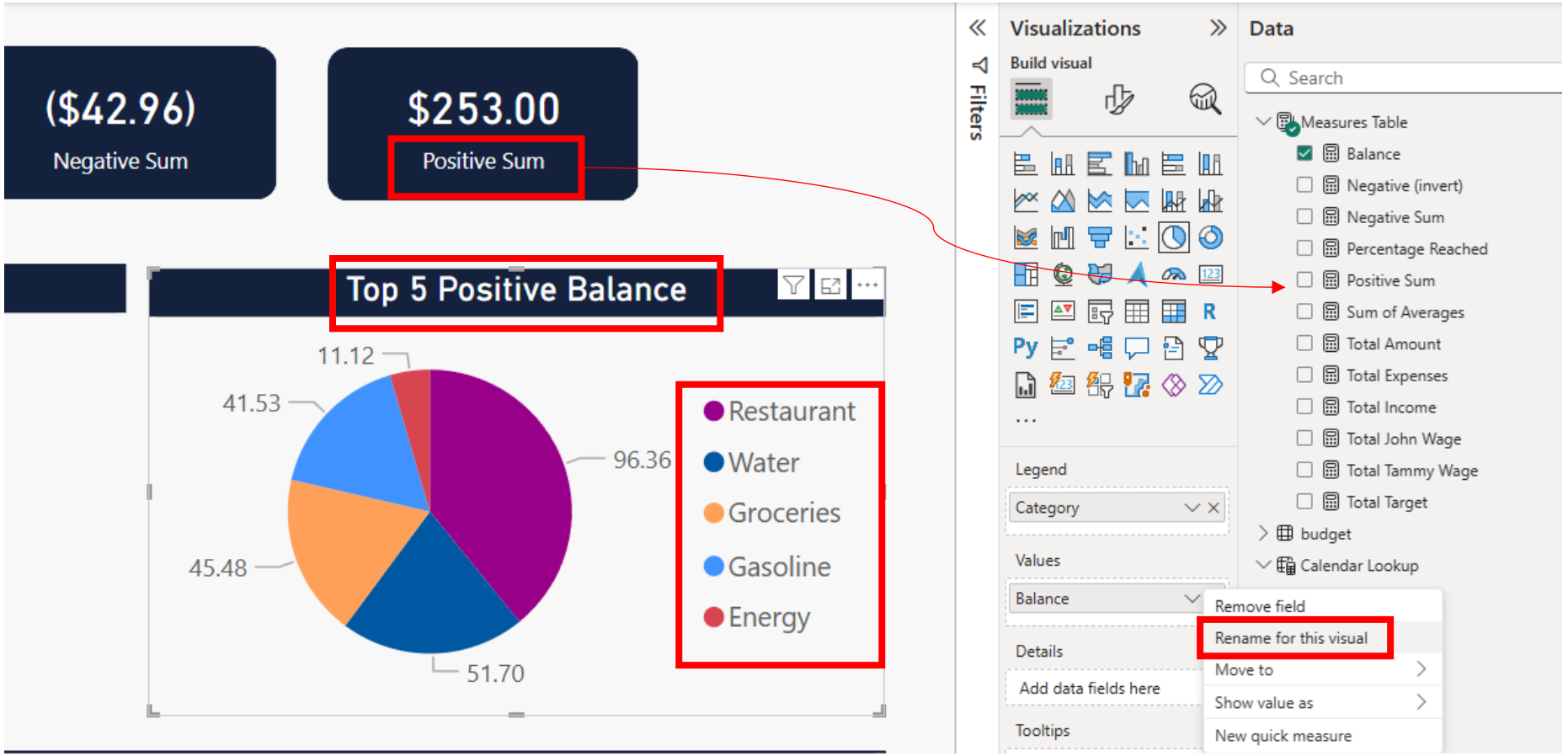
# Questionnaire

| Translation Level | Description | Translated Elements |
|---|---|---|
| 1) Which **languages & locales**? | Drives culture formats and total translation scope. | de-DE, en-GB |
| 2) **Default language** on open? | Sets initial view and fallback. | Default = de-DE |
| 3) How is language chosen: **manual slicer** or **auto** from browser/device? | Decides implementation (slicer/measure vs client culture). | Manual slicer |
| 4) Scope: translate **model captions/descriptions** (tables/columns/measures)? | Requires Tabular Editor translations; affects Field list/tooltips. | Yes, captions for users |
| 5) Scope: translate **visual texts** (titles, axis titles, tooltips, buttons, page titles)? | Determines number of translation keys/measures. | Titles + buttons + tooltips |
| 6) Any **data values** that need translation (categories, statuses, month names)? | May need lookup tables/field parameters and sorting. | "Accessories" → "Zubehör"; Month sorted by MonthNum |
| 7) **Formats**: currency, number, date/time, fiscal calendar & week start? | Controls dynamic format strings and Date table design. | EUR display; fiscal year starts Apr; week starts Mon |
| 8) **Page names & navigation** per language? | Affects bookmarks/buttons or dynamic text boxes. | "Overview" ↔ "Übersicht" |
| 9) **Images/icons** language-specific? | Asset management and conditional display. | help_EN.png vs help_DE.png |
| 10) Where do translations live & who maintains them? | Governance and ease of updates. | Marketing owns CSV on SharePoint; monthly updates |
| 11) **Fallback language** if a key is missing? | Prevents blank labels. | Fall back to EN |
| 12) Any **RLS** that could filter translation tables? | Avoids missing keys due to security filters. | RLS only on facts; translation tables unfiltered |
| 13) **Testing & acceptance criteria** per language/device? | Ensures quality before go-live. | "0 missing keys; currency/date correct; mobile checked" |
| 14) **Exports/subscriptions** (PDF/PPT/Excel): which language? | Impacts Service setup and workflows. | Subscriptions in user's slicer language |

# Translation options

| Option | Description | Pros | Cons | Best for |
|---|---|---|---|---|
| **1) Separate report per language** | One dataset, **duplicate the report** for each language. | Easiest to start | **Very high maintenance:** duplicate subscriptions/permissions, risk of drift | One-off PoC, ≤2 languages, rare updates. |
| **2) Separate dataset per language** | **Duplicate the semantic model** (and often ETL) per language; thin report bound per dataset. | Metadata & data naturally localized; works if sources are already split by country. | **High maintenance/cost** (refresh, ETL, governance); hard to keep logic in sync; not scalable. | Legacy setups with country-specific data pipelines. |
| **3) One report, duplicated pages + bookmarks** | Keep both language versions of each page; buttons/bookmarks toggle. | Designer-friendly; no DAX for labels. | **High maintenance** (every page twice), filtering/bookmark quirks, heavier file. | Small reports with lots of layout differences per language. |
| **4) One report + translation table (DAX)** ✅ | Single dataset/report. **Language slicer** + translation table drive all labels; field parameters or lookup table for data values. | **Best scalability/maintenance** | Some upfront setup; authors must wire titles via measures; large RTL layout changes are extra work. | Most production reports (multi-language, evolving content). |

# Understanding the Components of a Report

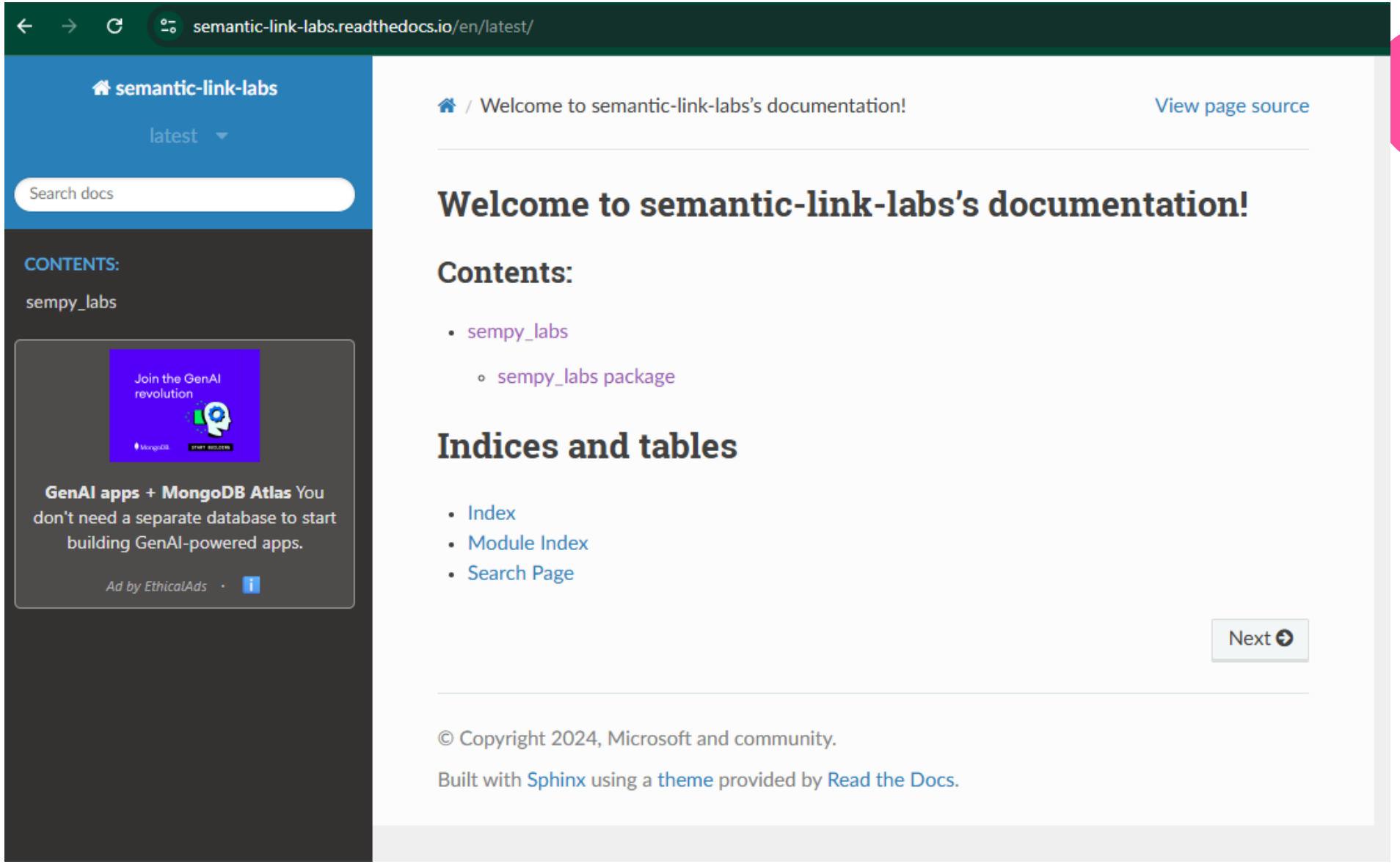



https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/Personal-Finance-Dashboard-Painel-de-Finan%C3%A7as-Pessoais/m-p/4671757

# Tools used: **Semantic Link Labs**

**Semantic Link Labs** is a Python library designed for use in Microsoft Fabric notebooks. This library extends the capabilities of Microsoft's **Semantic Link** offering additional functionalities to integrate and work with it.

- Export and Modify Semantic Models Programmatically
- Automated Translation of Model Metadata
- Bulk Metadata Editing
- Comparison Between Models
- Manage Role-Level Security
- Generate Documentation Automatically


semantic-link-labs.readthedocs.io/en/latest/
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© Copyright 2024, Microsoft and community.
Built with Sphinx using a theme provided by Read the Docs.

# Tools used: **Azure AI Translator**

Azure AI Translator is a cloud-based service that provides language translation, it uses Neural Machine Translation. Machine translation with REST API call. Supports over 90 languages and dialects.

**Pricing**:

- ➔ **Free tier** with up to **2 million characters** per month.
- ➔ Pay-as-you-go ranges from €8.85 (standard) to €35.39 (custom) per million characters.
- ➔ Other prices for document translation, training data, custom model hosting, high-volume usage, and annual container options are also available at https://azure.microsoft.com/en-us/pricing/details/cognitive-services/translator/

**sempy_labs.translate_semantic_model**(*dataset: str | UUID, languages: str | List[str], exclude_characters: str | None = None, workspace: str | UUID | None = None*) → DataFrame

Translates names, descriptions, display folders for all objects in a semantic model.

**Parameters:**

- **dataset** (*str | uuid.UUID*) – Name or ID of the semantic model.
- **languages** (*str, List[str]*) – The language code(s) in which to translate the semantic model.
- **exclude_characters** (*str*) – A string specifying characters which will be replaced by a space in the translation text when sent to the translation service.
- **workspace** (*str | uuid.UUID, default=None*) – The Fabric workspace name or ID. Defaults to None which resolves to the workspace of the attached lakehouse or if no lakehouse attached, resolves to the workspace of the notebook.

**Returns:** Shows a pandas dataframe which displays all of the translations in the semantic model.

**Return type:** pandas.DataFrame

Source: https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

## Translate a semantic model

```
import sempy_labs as labs

dataset = '' # Enter the name or ID of your semantic model
workspace = None # Enter the name or ID of the workspace in which the semantic model resides
languages = ['French', 'German', 'Chinese']

labs.translate_semantic_model(dataset=dataset, workspace=workspace, languages=languages)
```

Source: https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

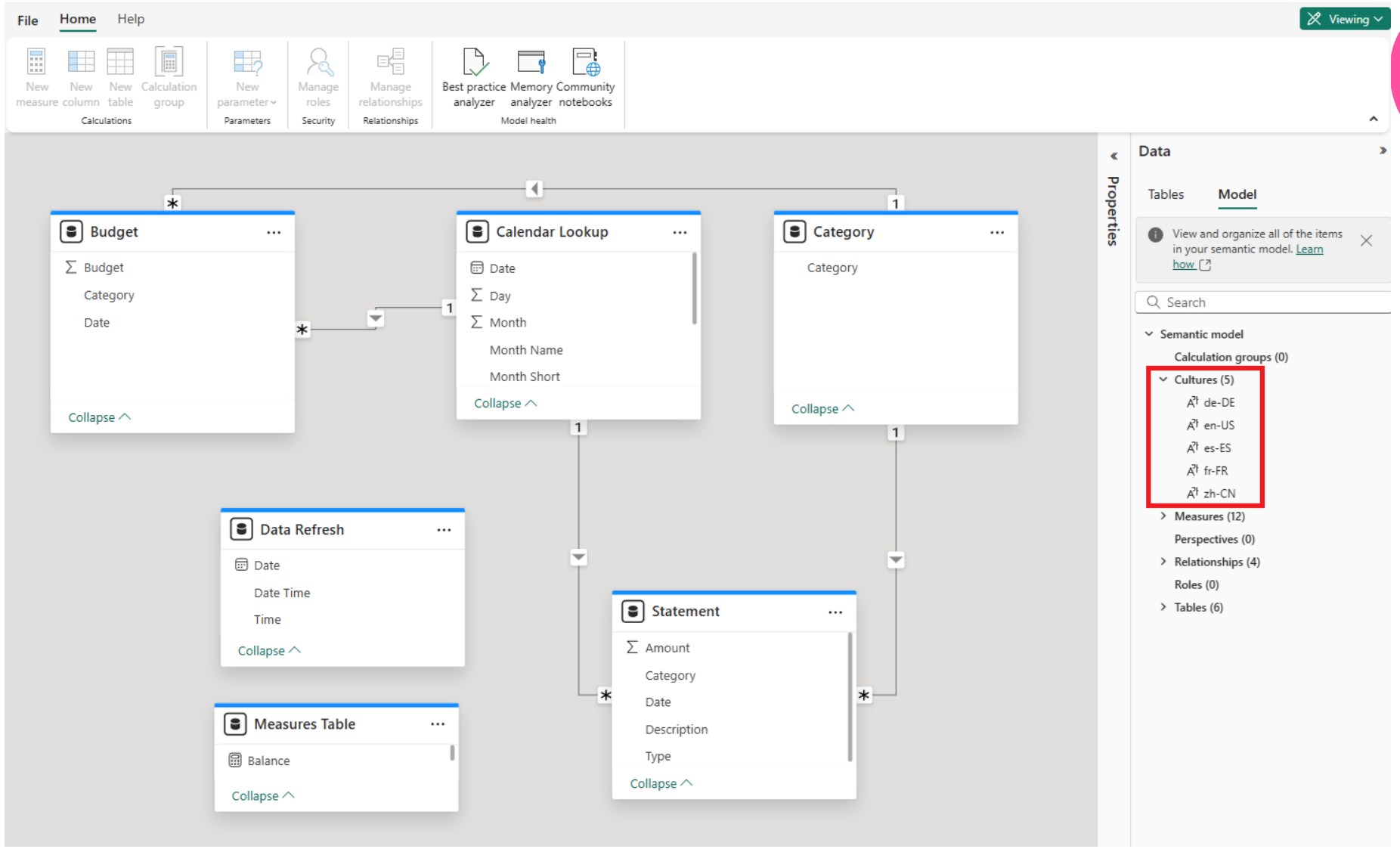
```python
import sempy_labs as labs

dataset = '79d1f506-3c87-49ee-84c0-95b72aa75743' # Enter the name or ID of your semantic model
workspace = None # Enter the name or ID of the workspace in which the semantic model resides
languages = ['French', 'German', 'Chinese', 'Spanish']

translations_df = labs.translate_semantic_model(dataset=dataset, workspace=workspace, languages=languages)
```

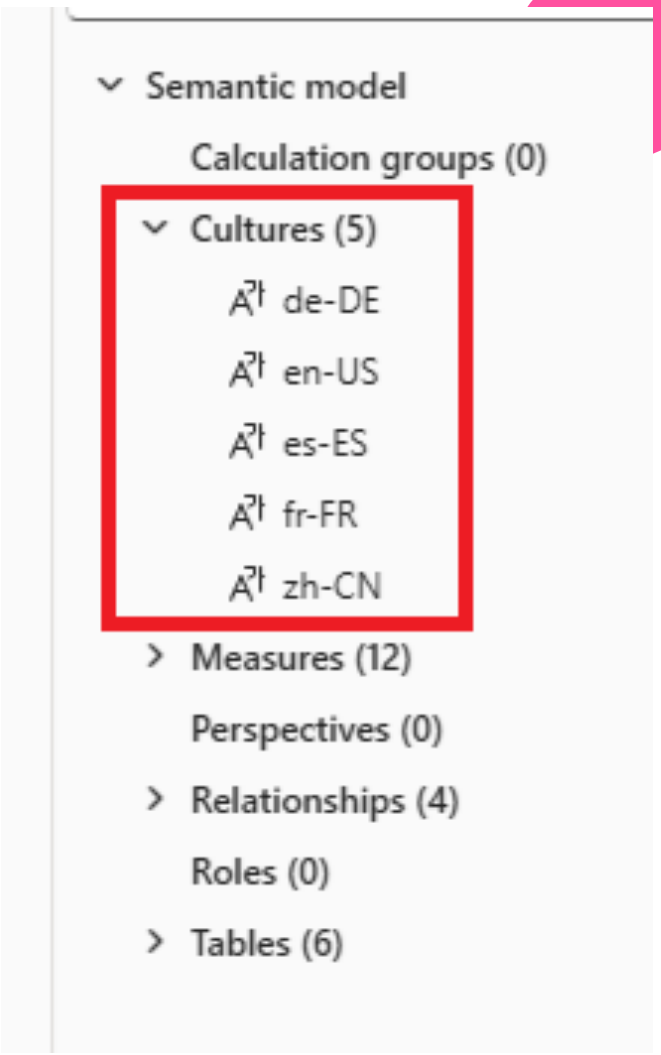
[*] 37 sec - Running PySpark (Python)

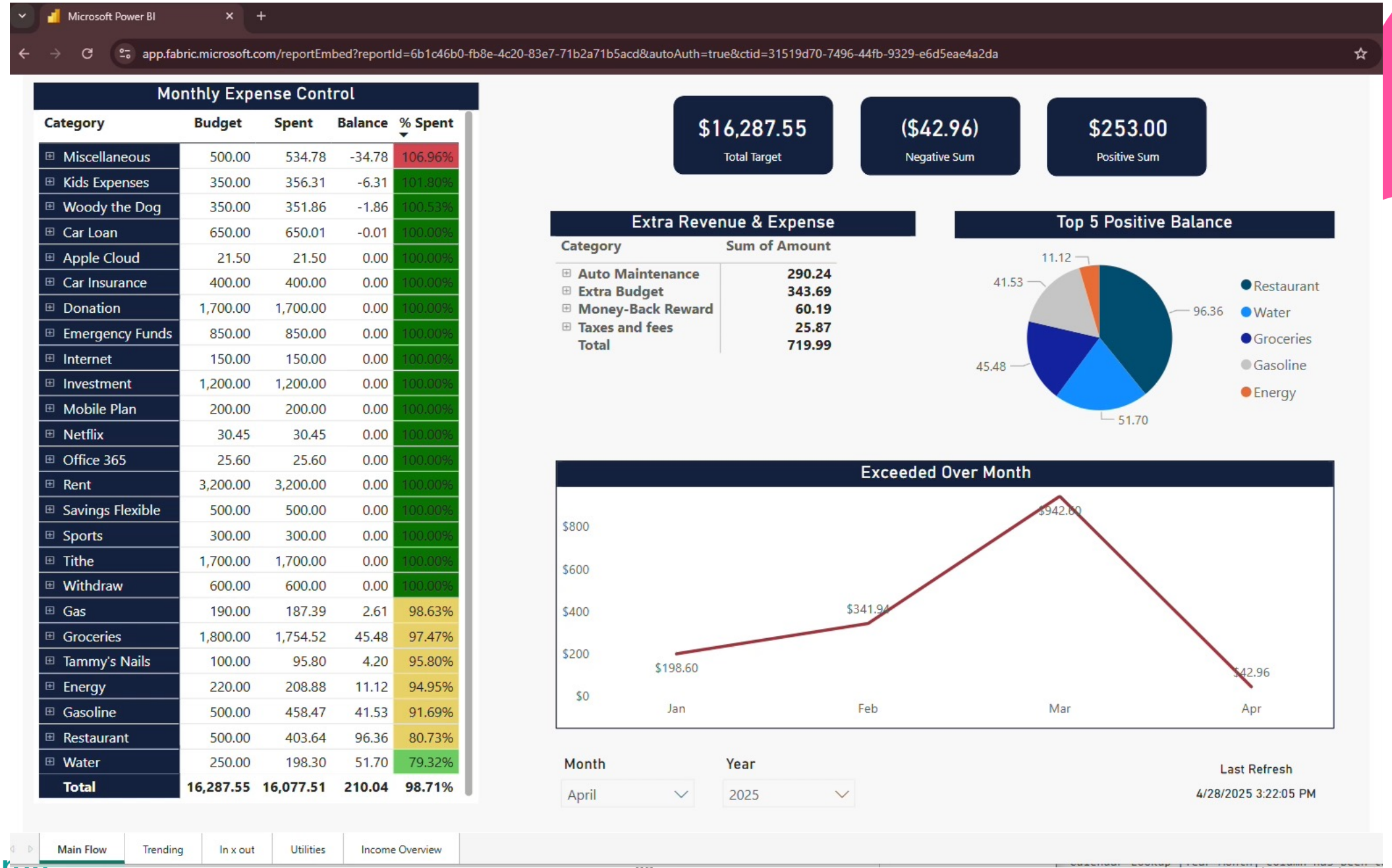
Spark jobs In progress (1) Resources Log


File
Home
Help
Viewing
New measure
New column
New table
Calculation group
Calculations
New parameter
Parameters
Manage roles
Security
Manage relationships
Relationships
Best practice analyzer
Memory analyzer
Community notebooks
Model health
Budget
Budget
Category
Date
Collapse
Calendar Lookup
Date
Day
Month
Month Name
Month Short
Collapse
Category
Category
Collapse
Data Refresh
Date
Date Time
Time
Collapse
Measures Table
Balance
Collapse
Statement
Amount
Category
Date
Description
Type
Collapse
Properties
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Semantic model
Calculation groups (0)
Cultures (5)
de-DE
en-US
es-ES
fr-FR
zh-CN
Measures (12)
Perspectives (0)
Relationships (4)
Roles (0)
Tables (6)

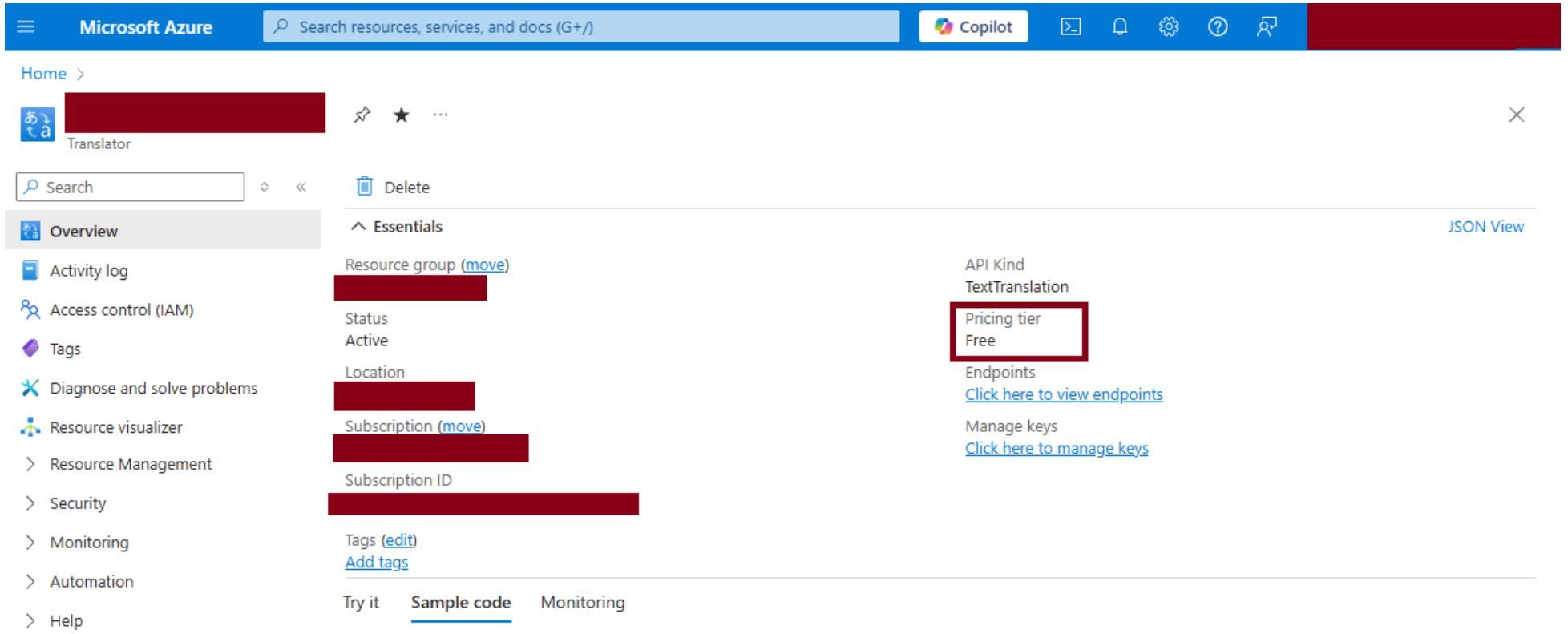
# 🧠 Translations become *Cultures* in the Semantic Model

- Each language is stored as a culture (e.g. fr-FR, es-ES, zh-CN)
- Cultures hold translations for table, column, measure metadata
- Power BI auto-selects culture based on the user's language
- Enables multilingual UX with a single report and dataset




Microsoft Power BI
app.fabric.microsoft.com/reportEmbed?reportId=6b1c46b0-fb8e-4c20-83e7-71b2a71b5acd&autoAuth=true&ctid=31519d70-7496-44fb-9329-e6d5eae4a2da
Monthly Expense Control
Category | Budget | Spent | Balance | % Spent
Miscellaneous 500.00 534.78 -34.78 106.96%
Kids Expenses 350.00 356.31 -6.31 101.80%
Woody the Dog 350.00 351.86 -1.86 100.53%
Car Loan 650.00 650.01 -0.01 100.00%
Apple Cloud 21.50 21.50 0.00 100.00%
Car Insurance 400.00 400.00 0.00 100.00%
Donation 1,700.00 1,700.00 0.00 100.00%
Emergency Funds 850.00 850.00 0.00 100.00%
Internet 150.00 150.00 0.00 100.00%
Investment 1,200.00 1,200.00 0.00 100.00%
Mobile Plan 200.00 200.00 0.00 100.00%
Netflix 30.45 30.45 0.00 100.00%
Office 365 25.60 25.60 0.00 100.00%
Rent 3,200.00 3,200.00 0.00 100.00%
Savings Flexible 500.00 500.00 0.00 100.00%
Sports 300.00 300.00 0.00 100.00%
Tithe 1,700.00 1,700.00 0.00 100.00%
Withdraw 600.00 600.00 0.00 100.00%
Gas 190.00 187.39 2.61 98.63%
Groceries 1,800.00 1,754.52 45.48 97.47%
Tammy's Nails 100.00 95.80 4.20 95.80%
Energy 220.00 208.88 11.12 94.95%
Gasoline 500.00 458.47 41.53 91.69%
Restaurant 500.00 403.64 96.36 80.73%
Water 250.00 198.30 51.70 79.32%
Total 16,287.55 16,077.51 210.04 98.71%
$16,287.55
Total Target
($42.96)
Negative Sum
$253.00
Positive Sum
Extra Revenue & Expense
Category | Sum of Amount
Auto Maintenance 290.24
Extra Budget 343.69
Money-Back Reward 60.19
Taxes and fees 25.87
Total 719.99
Top 5 Positive Balance
11.12
41.53
96.36
45.48
51.70
Restaurant
Water
Groceries
Gasoline
Energy
Exceeded Over Month
$800
$600
$400
$200
$0
$198.60
$341.94
$942.60
$42.96
Jan
Feb
Mar
Apr
Month
April
Year
2025
Last Refresh
4/28/2025 3:22:05 PM
Main Flow
Trending
In x out
Utilities
Income Overview

# Azure AI Translator



# How to use:

- Endpoint
- Key
- From language
- To language
- Body

Try it | **Sample code** | Monitoring

Translate text instantly or in batches across more than 100 languages, powered by the latest innovations in machine translation. For more information about the sample code you can view our GitHub. Learn more

C# | Go | Java | Node.js | **Python**

Copy sample code | Open REST API guide | View on Github

☐ Insert my keys and region in sample code.

```python
import requests, uuid, json

# Add your key and endpoint
key = "<your-translator-key>"
endpoint = "https://api.cognitive.microsofttranslator.com"

# location, also known as region.
# required if you're using a multi-service or regional (not global) resource. It can be found in the Azure portal o
location = "<YOUR-RESOURCE-LOCATION>"

path = '/translate'
constructed_url = endpoint + path

params = {
    'api-version': '3.0'
    'from': 'en'
    'to': ['fr', 'zu']
}

headers = {
    'Ocp-Apim-Subscription-Key': key
    # location required if you're using a multi-service or regional (not global) resource.
    'Ocp-Apim-Subscription-Region': location
    'Content-type': 'application/json'
    'X-ClientTraceId': str(uuid.uuid4())
}
```

```python
df_category = spark.sql("SELECT * FROM my_lakehouse.category")

# 1. Get all unique values from a table
categories = df_category.select("Category").distinct().dropna().collect()

# 2. Translate row by row
translated_zh = []
for row in categories:
    original = row["Category"]
    try:
        result = client.translate(
            body=[{"text": original}],
            from_language="en",
            to_language=["zh"]
        )
        translated_zh.append((original, result[0].translations[0].text))
    except Exception as e:
        translated_zh.append((original, f"ERROR: {str(e)}"))
```

✓ 9 sec - Command executed in 9 sec 688 ms by Samantha Cruz on 8:13:00 AM, 6/14/25

Table | + New chart

Table view

| | ABC Category | ABC Category_es | ABC Category_zh |
|---|---|---|---|
| 1 | Miscellaneous | Misceláneo | 杂项 |
| 2 | Savings Flexible | Ahorro Flexible | 灵活储蓄 |
| 3 | Water | Agua | 水 |
| 4 | Kids Expenses | Gastos de los niños | 儿童消费 |
| 5 | Gasoline | Gasolina | 汽油 |
| 6 | Car Insurance | Seguro de auto | 汽车保险 |
| 7 | Rent | Alquilar | 租金 |
| 8 | Energy | Energía | 能源 |
| 9 | Sports | Deportivo | 体育 |
| 10 | Money-Back ... | Recompensa de devolución de dinero | 退款奖励 |
| 11 | Extra Revenue | Ingresos adicionales | 额外收入 |
| 12 | Emergency Fu... | Fondos de Emergencia | 应急资金 |
| 13 | Groceries | Comestibles | 杂货 |
| 14 | Withdraw | Retirar | 撤回 |
| 15 | Auto Mainten... | Mantenimiento de automóviles | 汽车保养 |
| 16 | Internet | Internet | 互联网 |
| 17 | John Wage | John Wage | 约翰·瓦格 |

# Unpivot table

```python
from pyspark.sql.functions import lit

df = spark.sql("SELECT * FROM my_lakehouse.category")

langs = [("Category", "en"), ("Category_es", "es"), ("Category_zh", "zh")]

df_unpivoted = None
for col_name, lang in langs:
    temp = df.selectExpr(f"{col_name} as Category").withColumn("Language", lit(lang))
    df_unpivoted = temp if df_unpivoted is None else df_unpivoted.unionByName(temp)

display(df_unpivoted)
```

✓ 1 sec - Command executed in 1 sec 584 ms by Samantha Cruz on 9:27:14 AM, 6/14/25

Table | + New chart

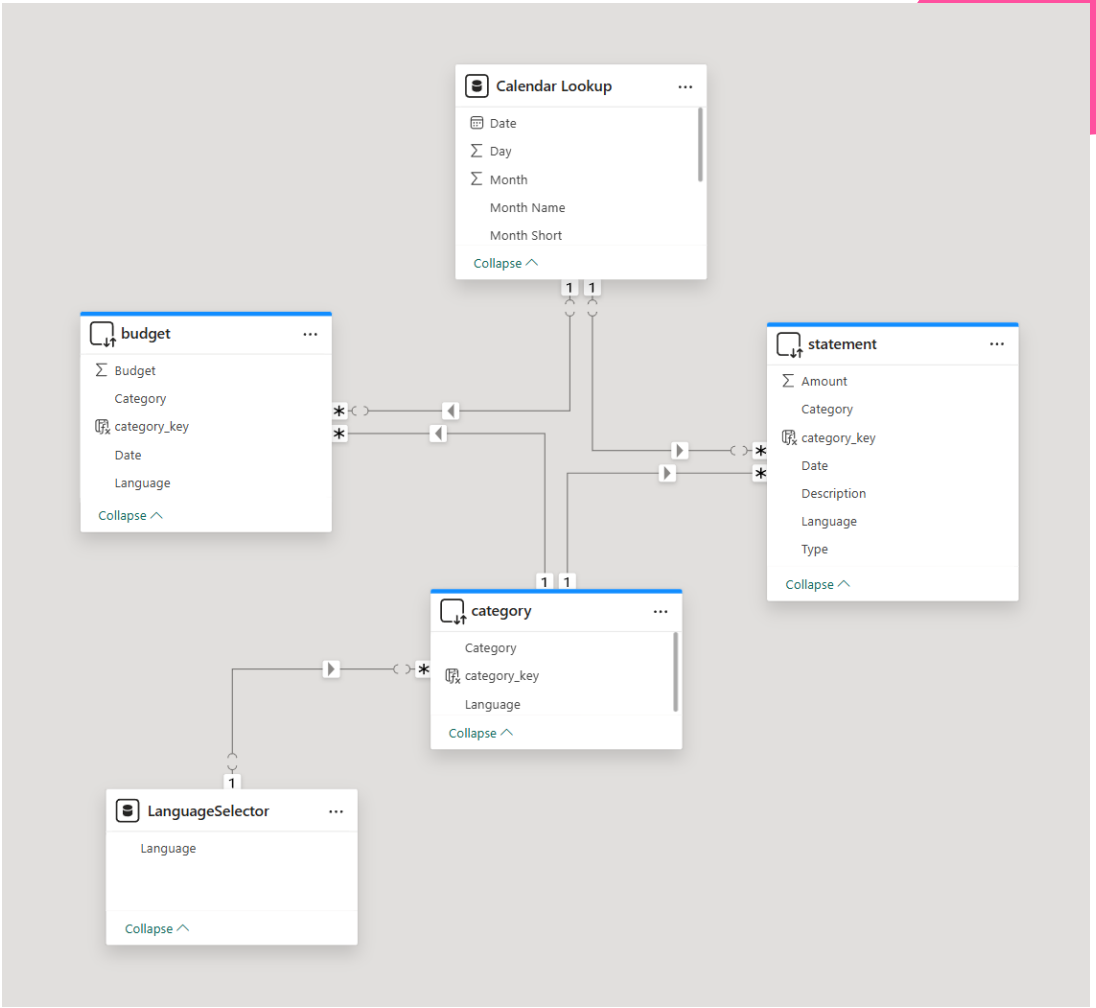
Table view

| | ABC Category | ABC Language |
|---|---|---|
| 59 | Plan Móvil | es |
| 60 | Gas | es |
| 61 | Inversión | es |
| 62 | Netflix (en ing... | es |
| 63 | Oficina 365 | es |
| 64 | Diezmo | es |
| 65 | 杂项 | zh |
| 66 | 灵活储蓄 | zh |
| 67 | 水 | zh |
| 68 | 儿童消费 | zh |
| 69 | 汽油 | zh |

# Language Dimension

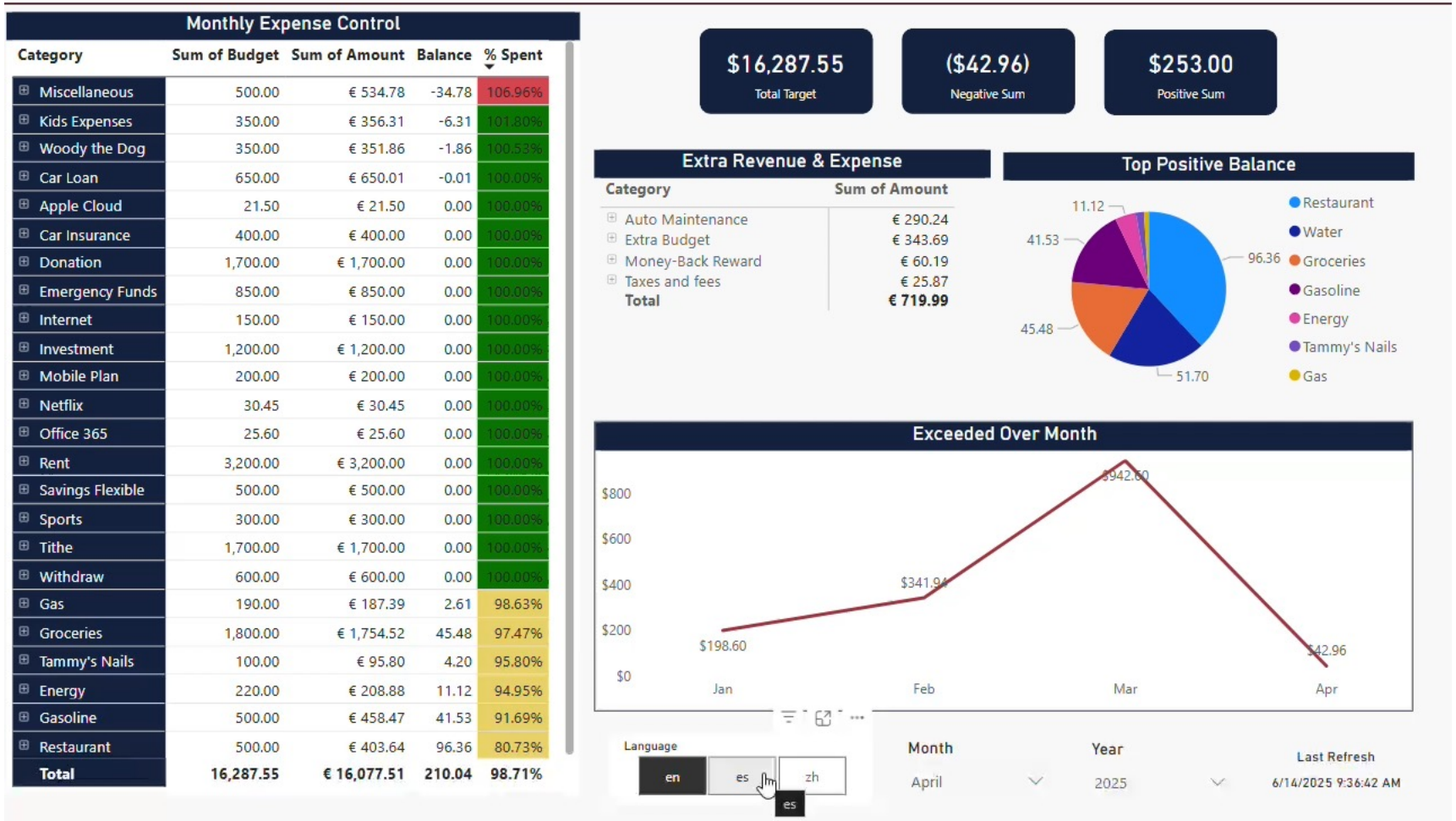
- A user picks a language from the slicer
- Power BI filters the category table by language
- That filtered translation is shown in visuals (matrix, tables, etc.)
- It works without switching pages or duplicating measures

Scales well with new languages (just add rows)
Avoids dynamic column/measure workarounds



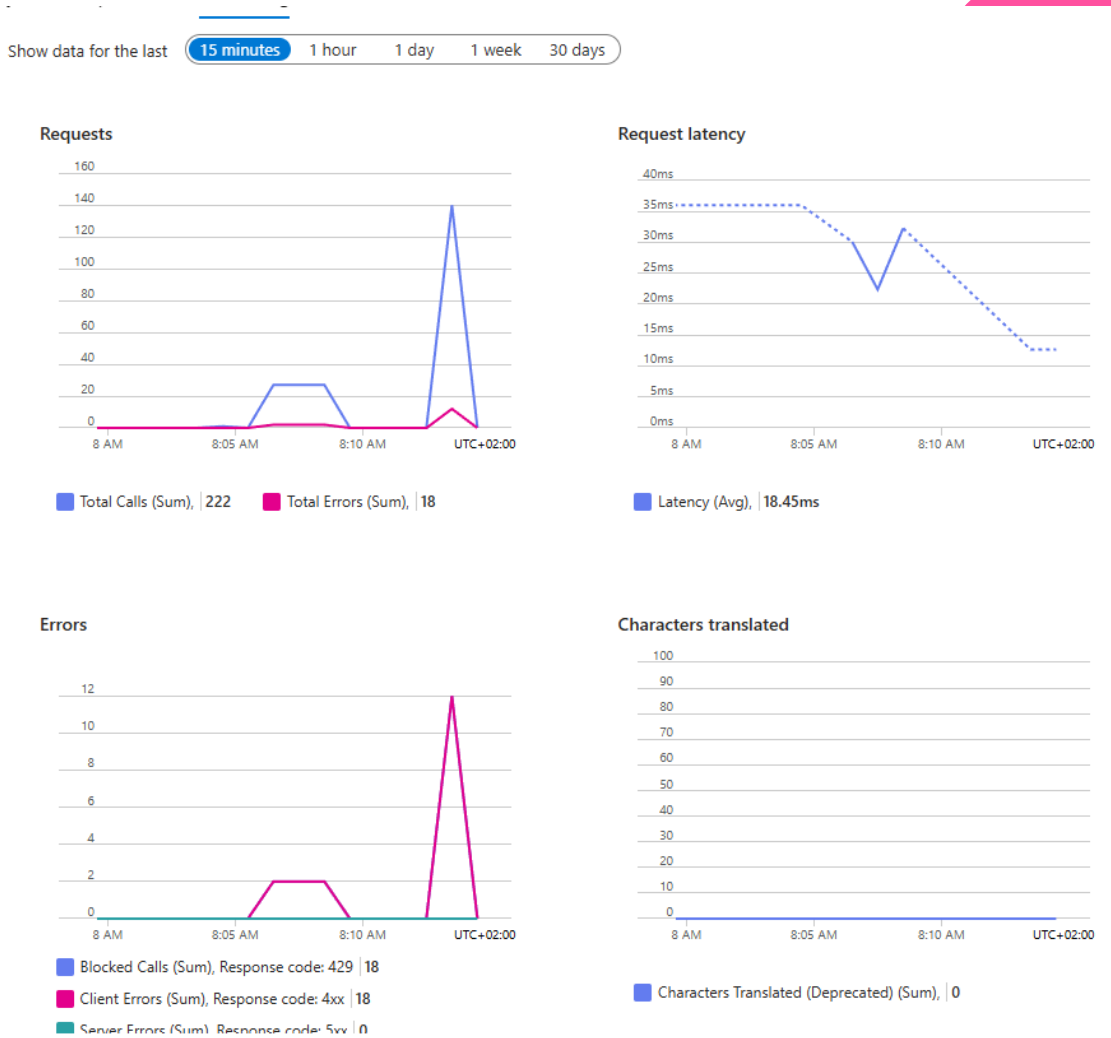



Confidential


Monthly Expense Control
Category | Sum of Budget | Sum of Amount | Balance | % Spent
Miscellaneous 500.00 € 534.78 -34.78 106.96%
Kids Expenses 350.00 € 356.31 -6.31 101.80%
Woody the Dog 350.00 € 351.86 -1.86 100.53%
Car Loan 650.00 € 650.01 -0.01 100.00%
Apple Cloud 21.50 € 21.50 0.00 100.00%
Car Insurance 400.00 € 400.00 0.00 100.00%
Donation 1,700.00 € 1,700.00 0.00 100.00%
Emergency Funds 850.00 € 850.00 0.00 100.00%
Internet 150.00 € 150.00 0.00 100.00%
Investment 1,200.00 € 1,200.00 0.00 100.00%
Mobile Plan 200.00 € 200.00 0.00 100.00%
Netflix 30.45 € 30.45 0.00 100.00%
Office 365 25.60 € 25.60 0.00 100.00%
Rent 3,200.00 € 3,200.00 0.00 100.00%
Savings Flexible 500.00 € 500.00 0.00 100.00%
Sports 300.00 € 300.00 0.00 100.00%
Tithe 1,700.00 € 1,700.00 0.00 100.00%
Withdraw 600.00 € 600.00 0.00 100.00%
Gas 190.00 € 187.39 2.61 98.63%
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Negative Sum
$253.00
Positive Sum
Extra Revenue & Expense
Category | Sum of Amount
Auto Maintenance € 290.24
Extra Budget € 343.69
Money-Back Reward € 60.19
Taxes and fees € 25.87
Total € 719.99
Top Positive Balance
11.12
41.53
96.36
45.48
51.70
Restaurant
Water
Groceries
Gasoline
Energy
Tammy's Nails
Gas
Exceeded Over Month
$800
$600
$400
$200
$0
$198.60
$341.94
$942.60
$42.96
Jan
Feb
Mar
Apr
Language
en
es
zh
es
Month
April
Year
2025
Last Refresh
6/14/2025 9:36:42 AM

# Monitoring

- Total Request Volume and Errors
- Latency and Performance Monitoring
- Client-Side Error Analysis (4xx)
- Monitoring Best Practices and Recommendations



# Best Practices

1. You save if you send larger texts

- Azure AI Translator is billed by **number of characters sent**, but:
- Each request has **network and processing overhead**.
- Sending **multiple words in one request** is more efficient than one request per word.
- **Batch translation is supported**: up to **100 items per request**, and up to **5,000 characters per item**.

# Best Practices

2. Alternatives to translating line by line:

- Group unique text values
- Batch translate multiple entries
- Concatenate short texts into one string

```
body = [{"text": val} for val in values]
response = client.translate(
    body=body,
    from_language="en",
    to_language=["es"]
)
```

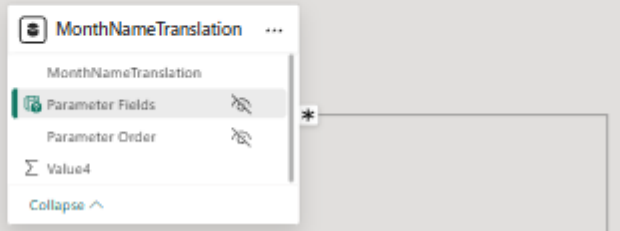
# Limitations

- Semantic Link Labs works only with semantic model metadata
- Azure AI Translator helps translate text, but does not apply it automatically
- Requires custom strategies to expose and manage translations
- Report layer elements (e.g. titles, buttons, text boxes) are not accessible via APIs
- Full automation of multilingual reports is not yet supported end-to-end
- Manual or embedded-layer solutions are needed for visual-level translations

# Parameterized Translation Tables

- Central **DimLanguage** table as language slicer
- Individual translation tables for each entity
- Composite key: [Entity ID] + [Language Code]
- Translated value stored in a Value or Valued column
- Relationships drive context filtering based on language

```
1 MonthNameTranslation = {
2     ("Mois", NAMEOF('DimCalendar'[MonthNameFR]), "FR", 0),
3     ("Monat", NAMEOF('DimCalendar'[MonthNameDE]), "DE", 1)
4 }
```

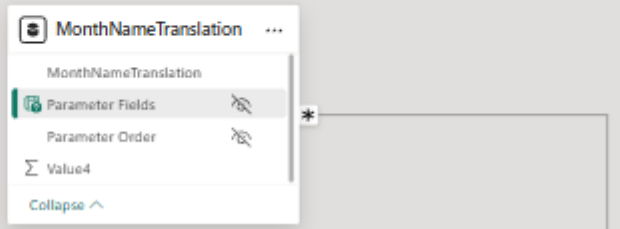


# Parameterized Translation Tables

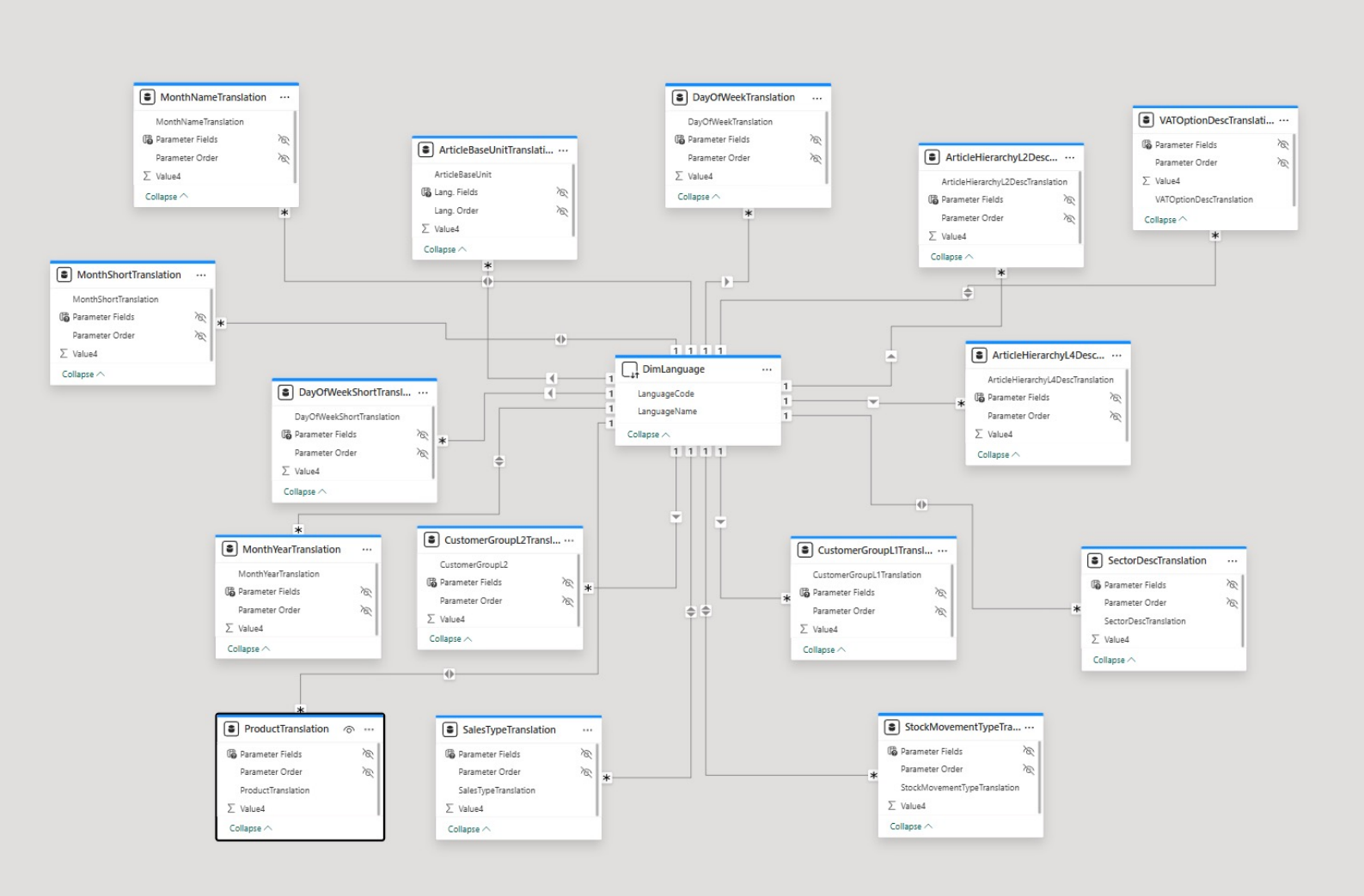
Benefits

- Centralized control of translations
- Scales across multiple entities and visuals
- Fully dynamic – responds to user language selection
- Clean separation between data and presentation layer
- Supports automation and governance

```
MonthNameTranslation = {
    ("Mois", NAMEOF('DimCalendar'[MonthNameFR]), "FR", 0),
    ("Monat", NAMEOF('DimCalendar'[MonthNameDE]), "DE", 1)
}
```



# Field Parameters



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